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Signals & Systems

Computer Assignment 4

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1 Laplace Transform

1.1 Introduction

In the first part, we have an RLC circuit with varying resistor. The transfer function changes due the value of the resistor(R). In the next part, we analyze an unity-feedback control system. Different K gives us different characteristics.

1.2 RLC Circuit

1.2.1 MATLAB Code

```
1 %Constants:
2 C = 1;
3 L = 1;
4 R = [0.5, 2, 5];
5 %Define Transfer Function
6 for i=1:3
7     sys(i) = tf(1, [1 , R(i),1]);
8 end
9 %Plot Bode Diagram
10 for i=1:3
11     figure;
12     bode(sys(i));
13     grid("on");
14 end
```

1.2.2 Result

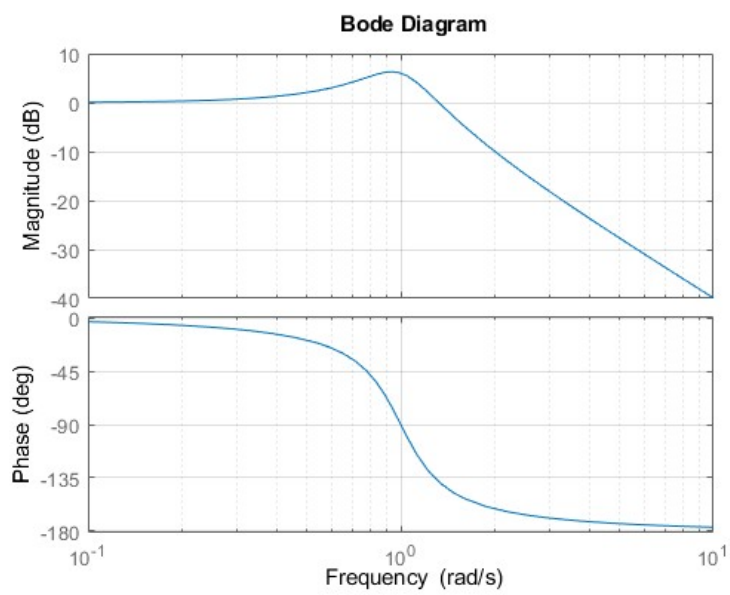
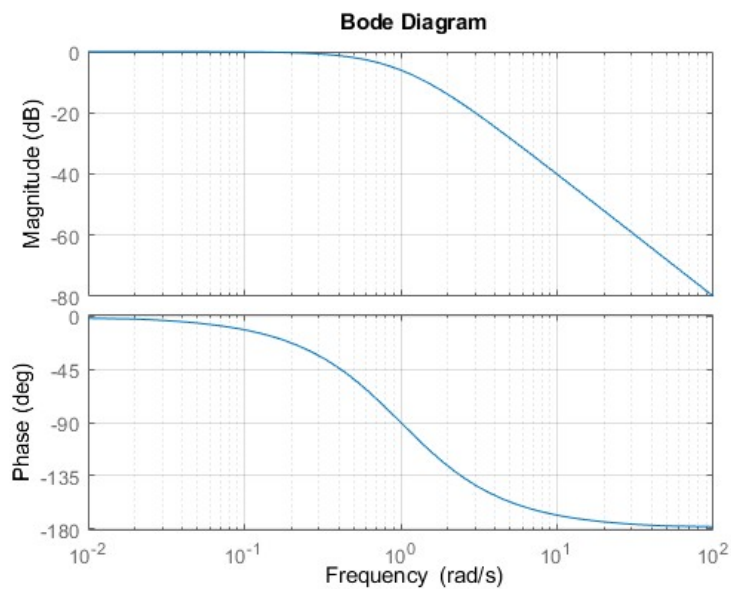
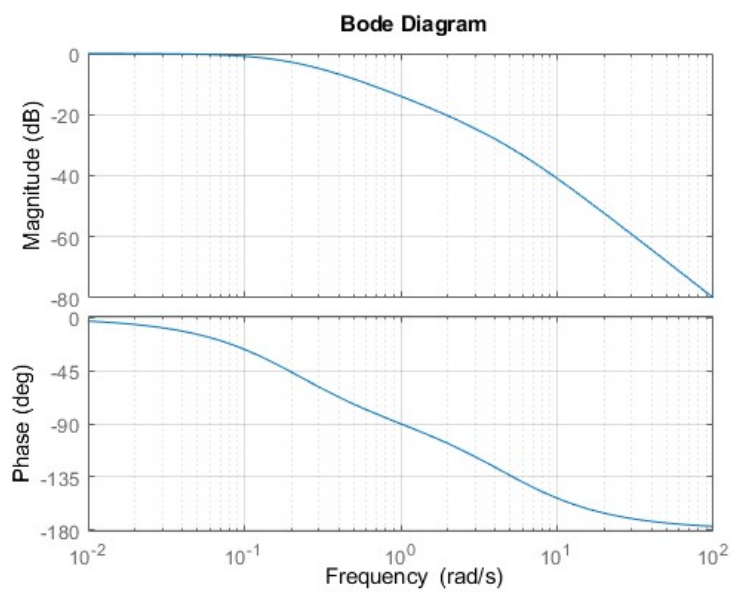


Figure 1: Transfer Function for $R = 0.5 \Omega$

Figure 2: Transfer Function for $R = 2\Omega$ Figure 3: Transfer Function for $R = 5\Omega$

As it is shown in Figure 1, 2 and 3, the filter is **Low-Pass**. The magnitude goes to zero as frequency increases. The cut-off frequency (and **Q**) decreases as we increase the resistor value. **Q** shows how wide our bandwidth is. It is a quality factor for the filter.

But for a small resistor, the characteristic is different. At frequency ω_o , magnitude rapidly increases and creates a peak (Figure 4).

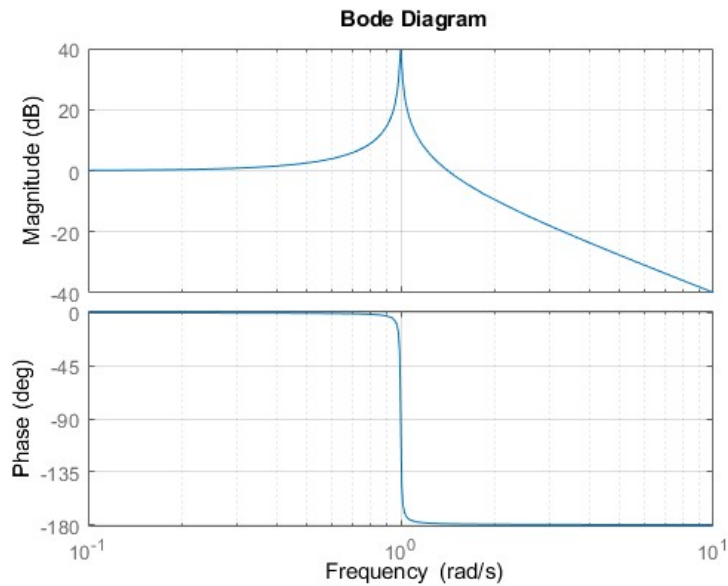


Figure 4: Transfer Function for $R = 0.01 \Omega$

1.3 Closed-loop Control System Stability

1.3.1 MATLAB Code

```

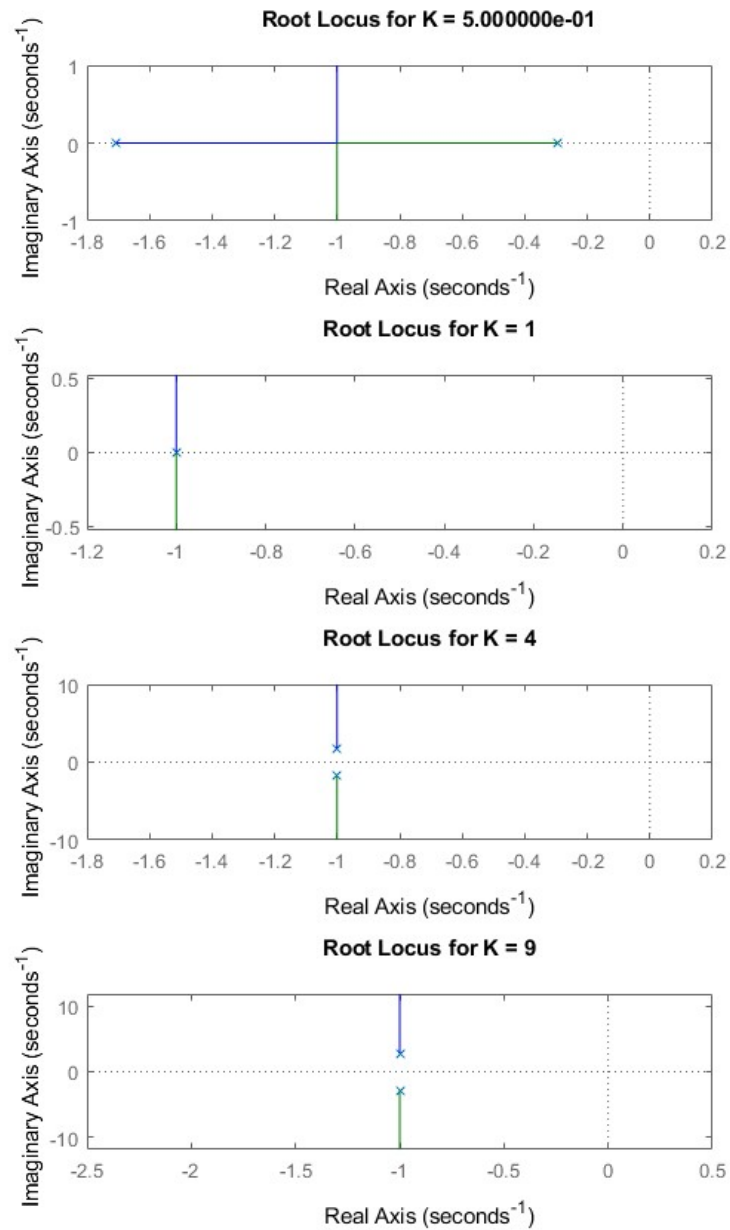
1 K = [0.5, 1, 4, 9];
2 %Feedback and rlocus plot
3 for i=1:length(K)
4     G(i) = tf(K(i), [1, 2, 0]);
5     unity = tf(1,1); %Unity filter
6     sys(i) = feedback(G(i), unity); %feedback
7     figure;
8     rlocus(sys(i));

```

```
9  end
10 %Bode plot
11 for i=1:length(K)
12     figure;
13     bode(sys(i));grid on;
14 end
```

1.3.2 Result

If roots are in the left-side plane, system is stable. Due to the Figure 5, they are all stable.

Figure 5: Root Locus plots for different K

There are three different type:

- Over-damped: Two roots are real and negative.
- Under-damped: Two roots are complex and have negative real part.
- Critically-damped: There is only one negative real root.

By this definition, $K=4$ and $K = 9$ are under-damped. $K=1$ is critically damped. $K = 0.5$ is over-damped. Bode diagram of each K are shown in Figure 6.

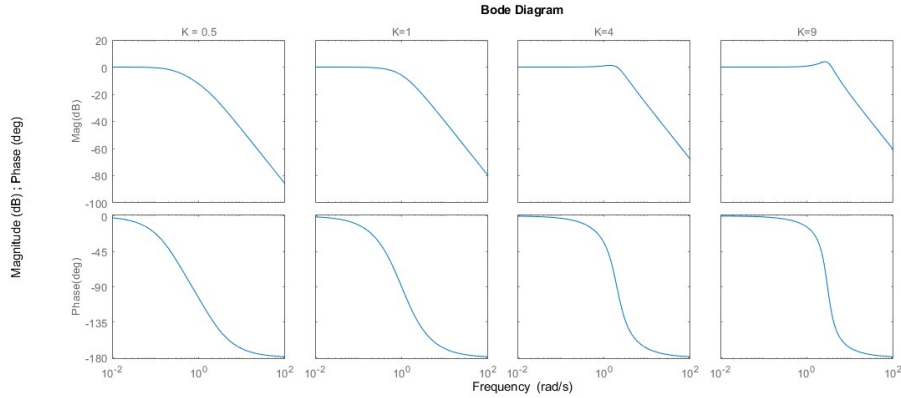


Figure 6: Bode plot for different K

2 Z-Transform

2.1 Moving Average

2.1.1 Theory

The N-point moving average filter is defined as:

$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[n-k] \quad (1)$$

We take the Z-transform of both sides of the equation.

$$Y(z) = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k} X(z) \quad (2)$$

$$H(z) = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k} = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}} \quad (3)$$

If you input an impulse $\delta[n]$ into the system, the coefficient of the sum dictate that output will be $\frac{1}{N}$ for $n = 0, \dots, N-1$ and 0 for everywhere else. Because this impulse response has a finite duration, it is by definition an FIR filter.

2.1.2 Code

```

1 N = [3,7,15];
2 figure;
3 for i=1:length(N)
4     n = N(i);
5     numerator = zeros(1, n+1);
6     denominator = zeros(1,n+1);
7     numerator(1) = 1./n; numerator(n+1) = -1./n;
8     denominator(1) = 1; denominator(2) = -1;
9
10    subplot(1,length(N),i);
11    zplane(numerator,denominator);
12 end

```

2.1.3 Result

In the Figure 7, zero-pole plot is shown. Zeros are distributed uniformly around the unit circle. We have a zero and a pole at $z = 1$. They perfectly cancel each other out. In reality, the system does not have a pole at $z = 1$.

Whenever the unit circle passes through a zero, the frequency response drops completely to zero. Because your zeros are at $\omega = \frac{2\pi k}{N}$, the filter will completely block those specific frequencies (and their harmonics).

The $(N-1)$ poles at $(0,0)$ do not change the magnitude of the frequency response because their distance to any point on the unit circle is always exactly 1. They only introduce a linear time delay to the signal.

Because the zero at $z = 1$ (which corresponds to DC, or $\omega = 0$) was canceled by the pole, there is no zero there to attenuate the signal. The magnitude response is at its absolute maximum at $\omega = 0$.

In the frequency domain, an N -point moving average filter produces a `sinc`-like shape (Take a look at Figure 8). Due to the figure, it is a low-

pass filter. Increasing N will produce more zeros in Figure 8 (The number of roots is $\lceil N/2 \rceil$).

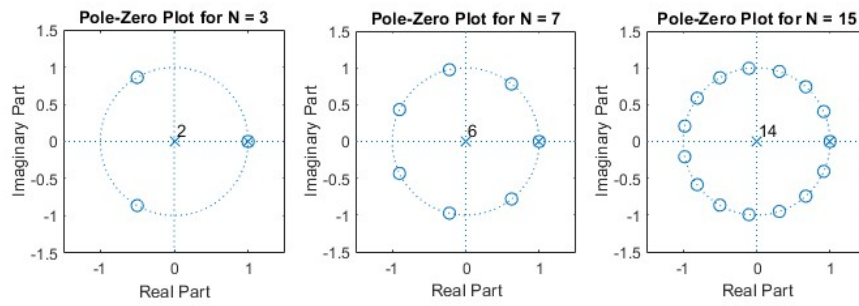


Figure 7: Zero-Pole plot for a moving average filter with different N

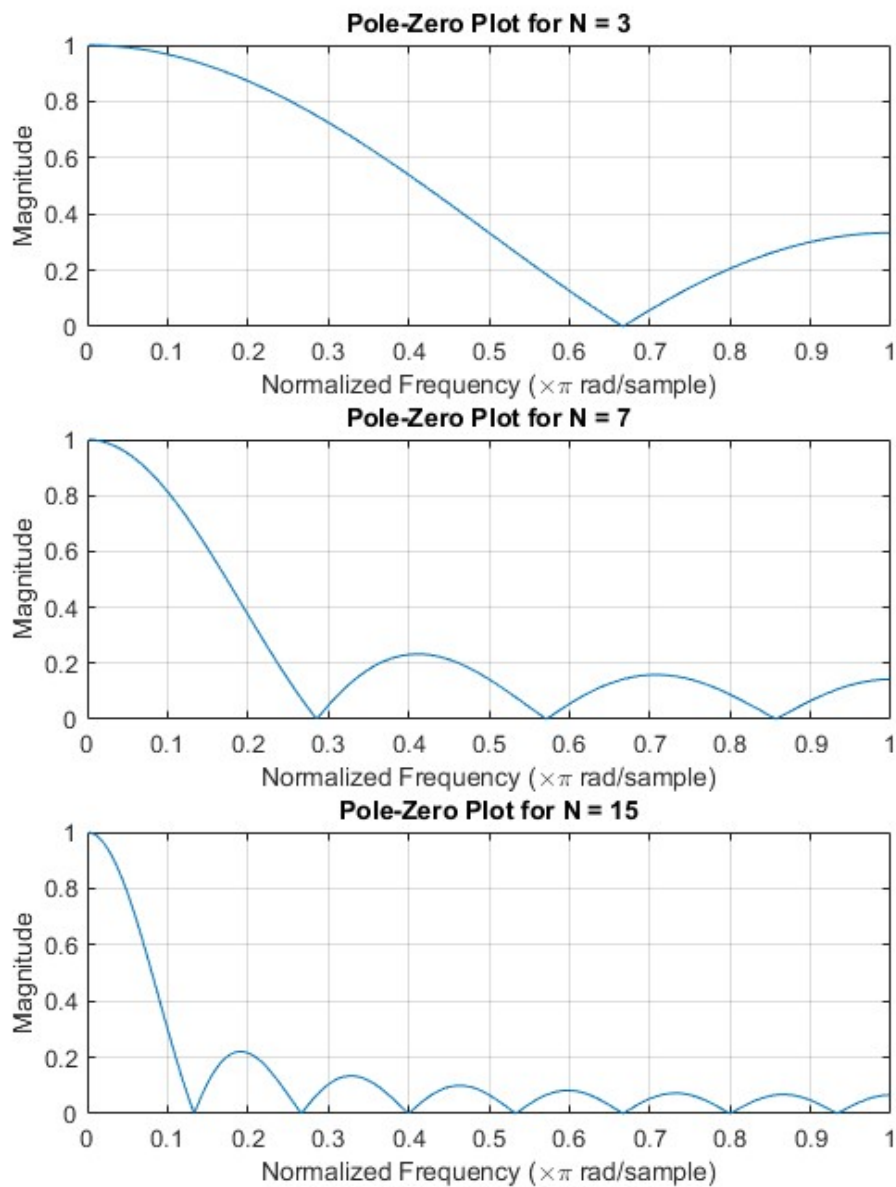


Figure 8: Frequency Response for a moving average filter with different N

2.1.4 Application

As it is shown in Figure 9, moving average reduces random noise.

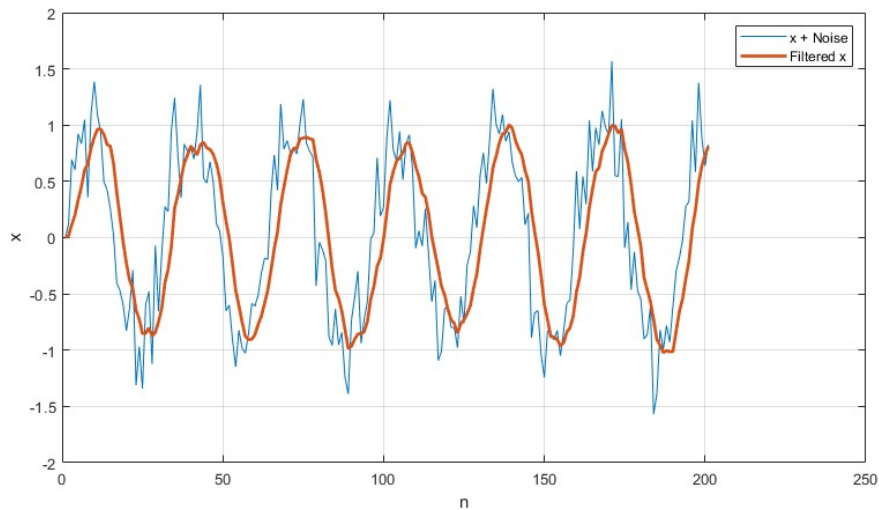


Figure 9: Moving Average reduces random noise.

2.2 Inverse Via Partial Fractions

For a system to be BIBO stability requires all poles to lie strictly inside the unit circle in the z -plane so, this system is BIBO due to the Figure 10.

2.2.1 Pole-Zero Plot

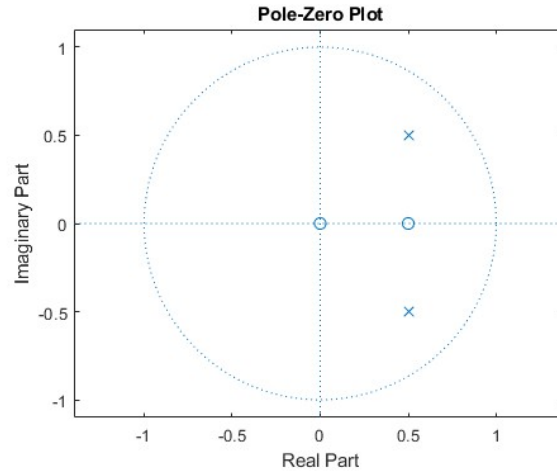


Figure 10: Zero-Pole plot of the transfer function.

After applying `iztrans` ,

$$h[n] = \frac{(-1)^n \cos\left(\frac{3\pi n}{4}\right)}{2^{n/2}} \quad (4)$$

As it is shown in Figure 11, sum over `n` would be finite so, the system is BIBO.

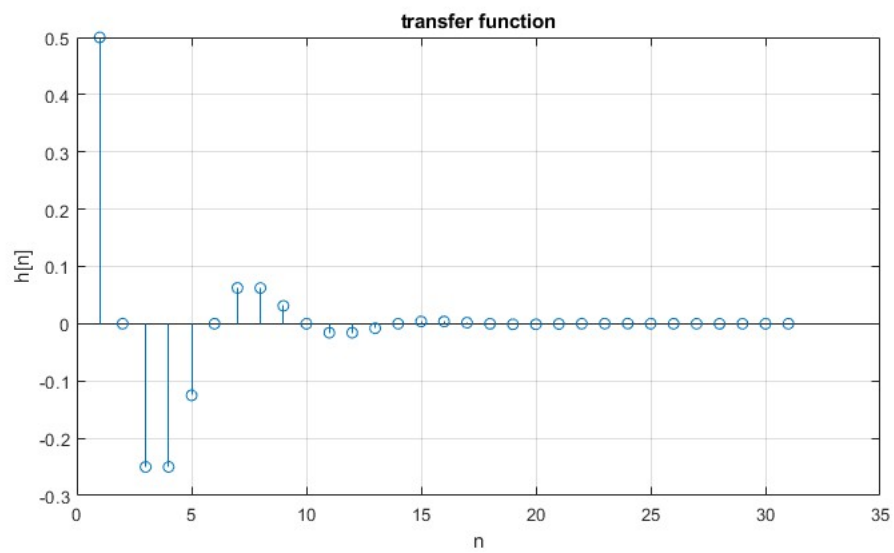


Figure 11